

- 6 (a) The banana plant, *Musa acuminata*, is a tall herbaceous plant with very large leaves.

Fig. 6.1 shows a banana plant.



Fig. 6.1

An investigation was carried out to measure the net carbon dioxide uptake by a banana plant at different light intensities.

Fig. 6.2 shows the results of the investigation.

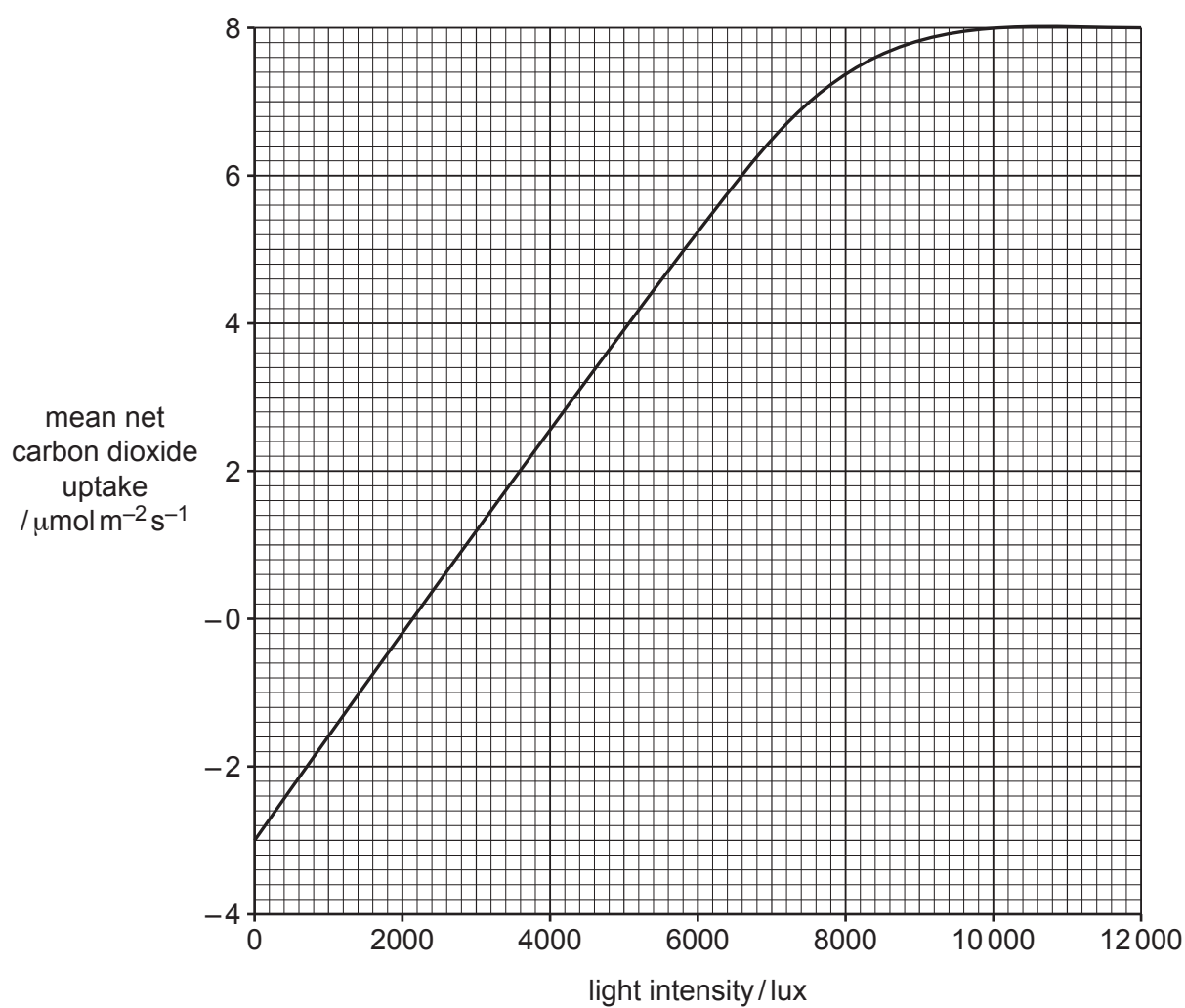


Fig. 6.2

- (i) With reference to Fig. 6.2, describe **and** explain the results at a light intensity of 1000 lux.

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..... [3]

- (ii) With reference to Fig. 6.2, describe **and** explain what can be concluded from the graph at light intensities of between 2000 lux and 7000 lux.

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..... [3]

- (iii) Explain why the rate of carbon dioxide uptake levels off as light intensity increases above 10 000 lux.

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..... [1]

- (b) Pitcher plants are carnivorous plants that trap and digest insects in a large modified leaf called a pitcher.

Fig. 6.3 shows a pale pitcher plant, *Sarracenia alata*.

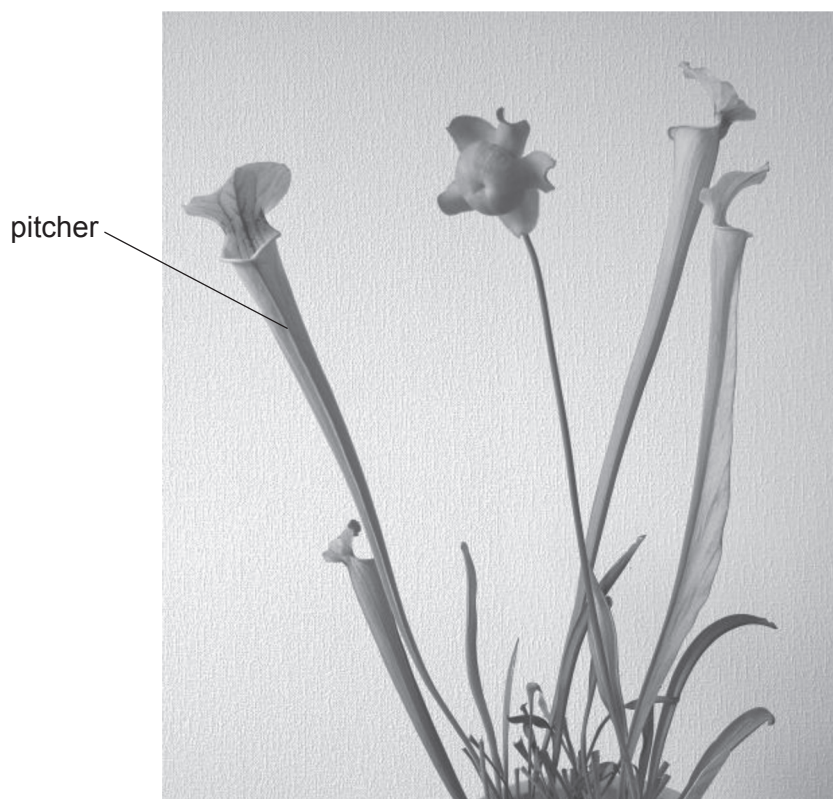


Fig. 6.3

- (i) Unlike the Venus fly trap, pitcher plants have no moving parts to trap insects. Insects are attracted to scent produced at the top of the pitcher. Once inside the pitcher, the insect slides down to the bottom and into a liquid containing digestive enzymes.

Suggest a type of enzyme found in the pitcher plant that is used to digest insects.

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- (ii) Pitcher plants grow in bogs and wet grassland where the soil has a low concentration of minerals such as nitrates. The plants obtain nitrates from the digestion of insects.

Calvin cycle intermediates are used to synthesise more complex compounds, some of which need nitrogen from nitrates.

Name **one** compound that needs nitrogen from nitrates **and** name the Calvin cycle intermediate from which it is synthesised.

compound that needs nitrogen

Calvin cycle intermediate

[2]

[Total: 10]

Question 7 starts on page 22.